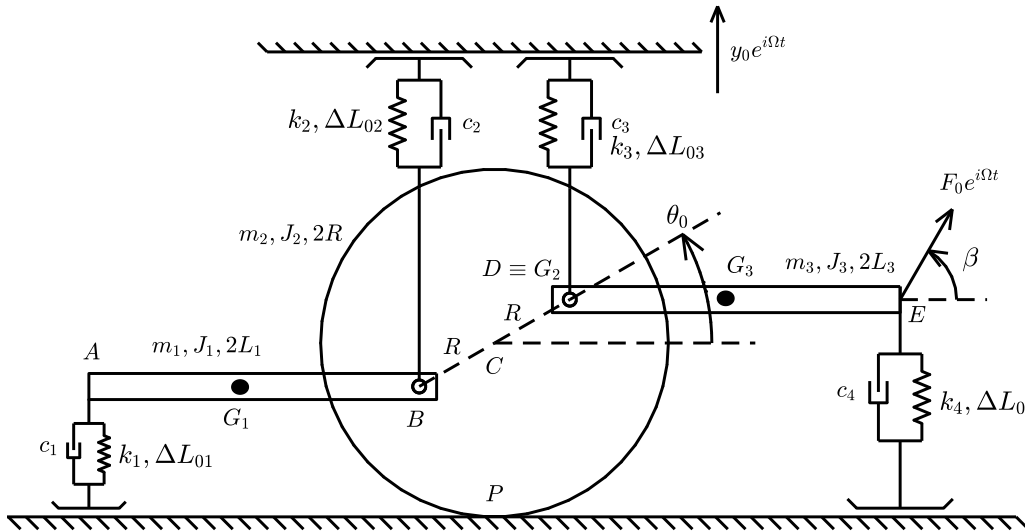


Dynamics of Mechanical Systems

January 09th, 2020 – standard problem (grading up to 30/30)



$m_1 = 6 \text{ kg}$	$k_1 = 5000 \text{ N/m}$
$m_2 = 1 \text{ kg}$	$k_2 = 9000 \text{ N/m}$
$m_3 = 4 \text{ kg}$	$k_3 = 13000 \text{ N/m}$
$J_1 = 0.08 \text{ kgm}^2$	$k_4 = 2000 \text{ N/m}$
$J_2 = 0.02 \text{ kgm}^2$	$c_1 = 2 \text{ Ns/m}$
$J_3 = 0.06 \text{ kgm}^2$	$c_2 = 10 \text{ Ns/m}$
$R = 0.1 \text{ m}$	$c_3 = 10 \text{ Ns/m}$
$L_1 = 0.2 \text{ m}$	$c_4 = 0.5 \text{ Ns/m}$
$L_3 = 0.2 \text{ m}$	$\theta_0 = \pi/6$
$\Delta L_{01} = -0.0059 \text{ m}$	$\beta = \pi/3$
$\Delta L_{02} = -0.005 \text{ m}$	
$\Delta L_{03} = -0.0035 \text{ m}$	
$\Delta L_{04} = -0.0098 \text{ m}$	

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. It is constituted by a disk with **non-homogeneous mass distribution** having radius $2R$ and **center of mass G_2** located at distance R from the geometric center C as shown in the figure. The disk rolls without sliding on the horizontal plane. Two rigid bars AB and DE are connected to the disk through hinges at points B and D , both at distance R from the disk's center. In the equilibrium position the angle between the BD direction and the horizontal axis is θ_0 .

The students are requested to:

- Write their surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet, according to the table shown below, leaving blank the field "Nr." Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;

Surname	Name	Matriculation number	SNxxx code	Nr.

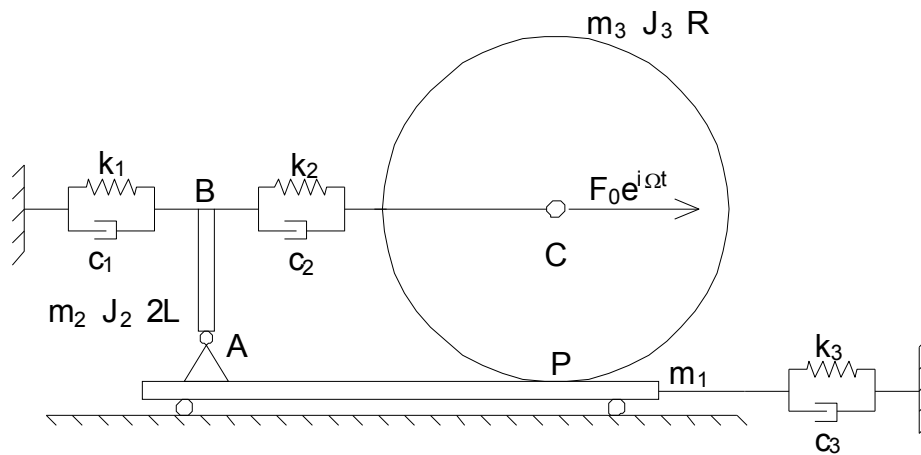
- Write the linearized equations of motion around the given equilibrium configuration;
- Compute the natural frequencies and modes of vibrations;
- Compute the Frequency Response Function (in the range $0 - 10 \text{ Hz}$, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the rotation of the DE rigid bar and the vertical component of the displacement of point G_1 . Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
- Compute the Frequency Response Function (in the range $0 - 10 \text{ Hz}$, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the rotation of the disk and the elastic and damping force transmitted by the spring/damper system (k_4, c_4). Save the graphical results as SNxxx3.fig and SNxxx4.fig
- Compute the Frequency Response Function (in the range $0 - 10 \text{ Hz}$, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$. Output: the dynamic component of the normal and tangential contact forces applied on the disk by the horizontal guide. Save the graphical result as SNxxx5.fig and SNxxx6.fig;
- Considering the loading case defined at the previous step with amplitude $F_0 = 10 \text{ N}$, plot the time history of the normal and tangential components of the contact forces for frequency $f = 2.5 \text{ Hz}$ ($dt = 0.001 \text{ s}$, consider three periods). Save the graphical result as SNxxx7.fig and SNxxx8.fig Compute the minimum value of the static friction coefficient f_s between the disk and horizontal guide needed to prevent the disk from slipping;
- Compute the static spring preload ΔL_{03} that keeps the mechanical system in the same static configuration, replacing the k_2 spring with another one that has the same undeformed length and new stiffness $k_{2n} = 4500 \text{ N/m}$.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

- Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
- Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
- Select the folder *Test January 09th 2020 - standard*
- Click on button Add and select Single document;
- Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
- Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
- Confirm the procedure clicking the button Publish.

Dynamics of Mechanical Systems

January 09th, 2020 – simplified problem (grading up to 22/30)



The mechanical system in the picture lays in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding along the rigid bar m_1 . The students are requested to:

1. Write their surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet, according to the table shown below, leaving blank the field “Nr.” Sheets with uncompleted personal data will not be considered. For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one;

Surname	Name	Matriculation number	SNxxx code	Nr.

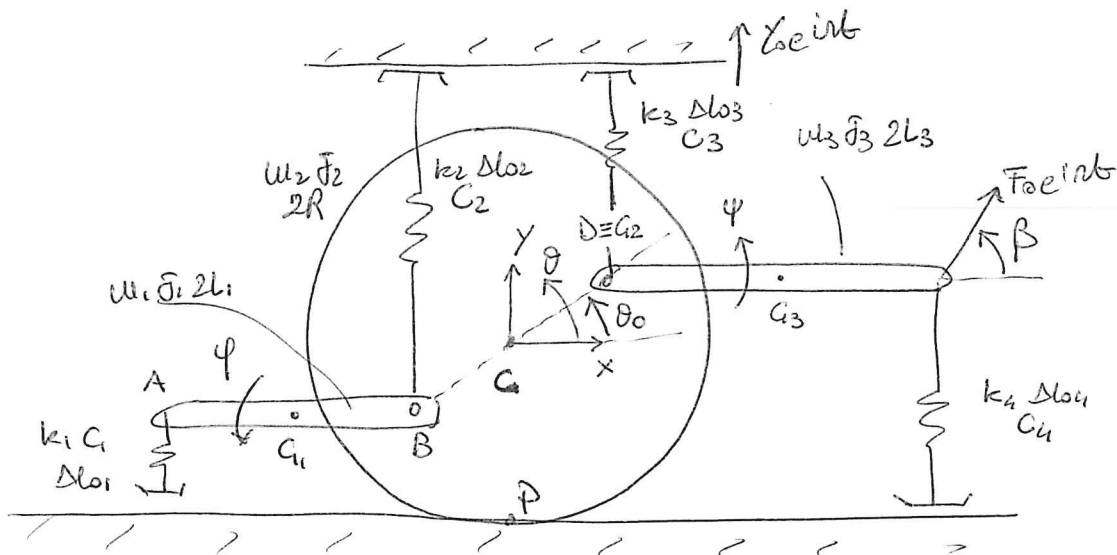
2. Write the linearized equations of motion around the given equilibrium configuration;
3. Compute the natural frequencies and modes of vibrations;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the horizontal harmonic force $F_0 e^{i\Omega t}$ at the point C. Output: the horizontal displacement of the slider m_1 and the horizontal component of the absolute displacement of point C. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the horizontal harmonic force $F_0 e^{i\Omega t}$ at the point C. Output: the elastic and damping force transmitted to the ground by the spring and damper system (k_1, c_1) . Save the graphical results as SNxxx3.fig.
6. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the horizontal harmonic force $F_0 e^{i\Omega t}$ at the point C. Output: the horizontal component of the constraint force in the hinge A. Save the graphical results as SNxxx4.fig.

$$\begin{aligned}
 m_1 &= 10 \text{ kg} & k_1 &= 1000 \text{ N/m} & L &= 0.25 \text{ m} \\
 m_2 &= 5 \text{ kg} & k_2 &= 500 \text{ N/m} & R &= 2L = 0.5 \text{ m} \\
 m_3 &= 10 \text{ kg} & k_3 &= 1000 \text{ N/m} \\
 J_2 &= 0.05 \text{ kgm}^2 & c_1 &= 1 \text{ Ns/m} \\
 J_3 &= 1.25 \text{ kgm}^2 & c_2 &= 0.1 \text{ Ns/m} \\
 & & c_3 &= 1 \text{ Ns/m}
 \end{aligned}$$

Note: Please collect the Matlab’s script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test January 09th 2020 - simplified*
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the “Title” field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.

WRITTEN TEST JANUARY 09th 2020
- standard problem -

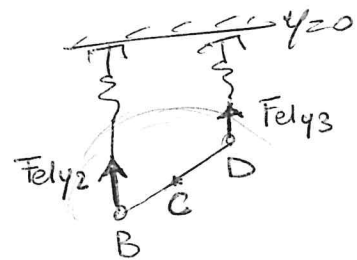


- 3 rigid bodies $\rightarrow$ 9 dof
- 2 hinges + rolling without sliding $\rightarrow$ 6 doc. $\rightarrow$ 3 dof

$$\underline{x} = \{\varphi \quad \theta \quad \psi\}^T \quad \underline{x}_0 = \{\emptyset \quad \emptyset_0 \quad \emptyset\}^T \quad \text{all absolute rotations}$$

- Simplified approach applying the equivalent forces F_{ely2} F_{ely3} at points D and B, and setting $y=0$

$$\begin{cases} F_{ely2} = (k_2 + i\omega C_2) y_0 \text{ eint} \\ F_{ely3} = (k_3 + i\omega C_3) y_0 \text{ eint} \end{cases}$$



$$T = \frac{1}{2} \omega_1 \dot{x}_{G1}^2 + \frac{1}{2} \omega_1 \dot{y}_{G1}^2 + \frac{1}{2} \bar{J}_1 \dot{\varphi}^2 + \frac{1}{2} \omega_2 \dot{x}_{G2}^2 + \frac{1}{2} \omega_2 \dot{y}_{G2}^2 + \frac{1}{2} \bar{J}_2 \dot{\theta}^2 + \frac{1}{2} \omega_3 \dot{x}_{G3}^2 + \frac{1}{2} \omega_3 \dot{y}_{G3}^2 + \frac{1}{2} \bar{J}_3 \dot{\psi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^4 C_i \dot{\Delta L}_i^2 ; \quad V = \frac{1}{2} \sum_{i=1,4} k_i \Delta L_i^2 + \sum_{i=1}^3 \omega_i g h_{G1}$$

$$\delta W = F_{ely2} \delta y_B + F_{ely3} \delta y_3 + F_{\phi x} \delta x_E + F_{\phi y} \delta y_E \quad \begin{aligned} F_{\phi x} &= F_0 \cos \beta \\ F_{\phi y} &= F_0 \sin \beta \end{aligned}$$

$$x_c = -2R\theta$$

$$x_B = -2R\theta - R\cos\theta$$

$$y_B = -R\sin\theta$$

$$x_{q1} = -2R\theta - R\cos\theta - L_1\cos\phi$$

$$y_{q1} = -R\sin\theta - L_1\sin\phi$$

$$y_A = -R\sin\theta - 2L_1\sin\phi$$

$$x_{q2} = -2R\theta + R\cos\theta$$

$$y_{q2} = R\sin\theta$$

$$x_{q3} = -2R\theta + R\cos\theta + L_3\cos\psi$$

$$y_{q3} = R\sin\theta + L_3\sin\psi$$

$$y_E = R\sin\theta + 2L_3\sin\psi$$

$$x_E = -2R\theta + R\cos\theta + 2L_3\cos\psi$$

$$\frac{\partial x_{q1}}{\partial \phi} = L_1\sin\phi \quad \frac{\partial x_{q1}}{\partial \theta} = -2R + R\sin\theta \quad \frac{\partial x_{q1}}{\partial \psi} = 0$$

$$\frac{\partial y_{q1}}{\partial \phi} = -L_1\cos\phi \quad \frac{\partial y_{q1}}{\partial \theta} = -R\cos\theta \quad \frac{\partial y_{q1}}{\partial \psi} = 0$$

$$\frac{\partial x_{q2}}{\partial \theta} = -2R - R\sin\theta \quad \frac{\partial x_{q2}}{\partial \phi} = \frac{\partial x_{q2}}{\partial \psi} = 0$$

$$\frac{\partial y_{q2}}{\partial \theta} = R\cos\theta \quad \frac{\partial y_{q2}}{\partial \phi} = \frac{\partial y_{q2}}{\partial \psi} = 0$$

$$\frac{\partial x_{q3}}{\partial \phi} = 0 \quad \frac{\partial x_{q3}}{\partial \theta} = -2R - R\sin\theta \quad \frac{\partial x_{q3}}{\partial \psi} = -L_3\sin\psi$$

$$\frac{\partial y_{q3}}{\partial \phi} = 0 \quad \frac{\partial y_{q3}}{\partial \theta} = R\cos\theta \quad \frac{\partial y_{q3}}{\partial \psi} = L_3\cos\psi$$

$$\frac{\partial x_E}{\partial \phi} = 0 \quad \frac{\partial x_E}{\partial \theta} = -2R - R\sin\theta \quad \frac{\partial x_E}{\partial \psi} = -2L_3\sin\psi$$

$$\frac{\partial y_E}{\partial \phi} = 0 \quad \frac{\partial y_E}{\partial \theta} = R\cos\theta \quad \frac{\partial y_E}{\partial \psi} = 2L_3\cos\psi$$

$$\Delta L_1 = y_A - y_{A\phi} \rightarrow \frac{\partial \Delta L_1}{\partial \phi} = -2L_1\cos\phi \quad \frac{\partial \Delta L_1}{\partial \theta} = -R\cos\theta \quad \frac{\partial \Delta L_1}{\partial \psi} = 0$$

$$\Delta L_2 = -y_B + y_{B\phi} \rightarrow \frac{\partial \Delta L_2}{\partial \phi} = 0 \quad \frac{\partial \Delta L_2}{\partial \theta} = R\cos\theta \quad \frac{\partial \Delta L_2}{\partial \psi} = 0$$

$$\Delta L_3 = -y_D + y_{D\phi} \rightarrow \frac{\partial \Delta L_3}{\partial \phi} = 0 \quad \frac{\partial \Delta L_3}{\partial \theta} = -R\cos\theta \quad \frac{\partial \Delta L_3}{\partial \psi} = 0$$

$$\Delta L_4 = y_E - y_{E\phi} \rightarrow \frac{\partial \Delta L_4}{\partial \phi} = 0 \quad \frac{\partial \Delta L_4}{\partial \theta} = R\cos\theta \quad \frac{\partial \Delta L_4}{\partial \psi} = 2L_3\cos\psi$$

$$[\Delta u] = \begin{bmatrix} 0 & -2R + R \sin \theta_0 & 0 \\ -L_1 & -R \cos \theta_0 & 0 \\ 1 & 0 & 0 \\ 0 & -2R - R \sin \theta_0 & 0 \\ 0 & R \cos \theta_0 & 0 \\ 0 & 1 & 0 \\ 0 & -2R - R \sin \theta_0 & 0 \\ 0 & R \cos \theta_0 & L_3 \\ 0 & 0 & 1 \end{bmatrix} \quad [\Delta u] = \begin{bmatrix} -2L_1 & -R \cos \theta_0 & 0 \\ 0 & R \cos \theta_0 & 0 \\ 0 & -R \cos \theta_0 & 0 \\ 0 & R \cos \theta_0 & 2L_3 \end{bmatrix}$$

$$[\Delta c] = [\Delta u]$$

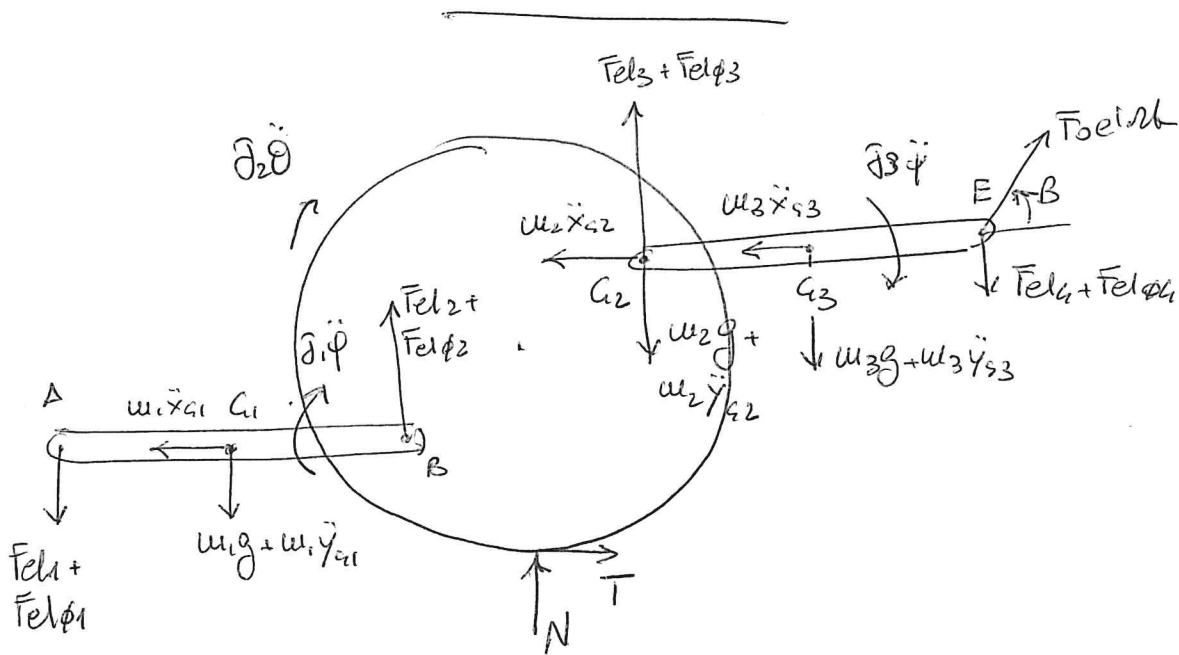
$$[\Delta d] = \begin{bmatrix} 0 & -R \cos \theta_0 & 0 \\ 0 & R \cos \theta_0 & 0 \\ 0 & -2R - R \sin \theta_0 & 0 \\ 0 & R \cos \theta_0 & 2L_3 \end{bmatrix}$$

$$\begin{aligned} \frac{\partial^2 \Delta u_1}{\partial \varphi^2} &= 2L_1 \sin \varphi & \frac{\partial^2 \Delta u_1}{\partial \theta^2} &= R \sin \theta \\ \frac{\partial^2 \Delta u_2}{\partial \theta^2} &= -R \sin \theta \\ \frac{\partial^2 \Delta u_3}{\partial \theta^2} &= R \sin \theta \\ \frac{\partial^2 \Delta u_4}{\partial \theta^2} &= -R \sin \theta & \frac{\partial^2 \Delta u_4}{\partial \varphi^2} &= -2L_3 \sin \varphi \end{aligned} \quad \left| \begin{aligned} \frac{\partial^2 \Delta u_1}{\partial \varphi^2} &= L_1 \sin \varphi & \frac{\partial^2 \Delta u_1}{\partial \theta^2} &= R \sin \theta \\ \frac{\partial^2 \Delta u_2}{\partial \theta^2} &= -R \sin \theta \\ \frac{\partial^2 \Delta u_3}{\partial \theta^2} &= R \sin \theta & \frac{\partial^2 \Delta u_3}{\partial \varphi^2} &= -L_3 \sin \varphi \end{aligned} \right.$$

$$[K_{EL, \Pi 1}] = k_1 \Delta l_1 \begin{bmatrix} 0 & 0 & 0 \\ 0 & R \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [K_{EL, \Pi 2}] = k_2 \Delta l_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & -R \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[K_{EL, \Pi 3}] = k_3 \Delta l_3 \begin{bmatrix} 0 & 0 & 0 \\ 0 & R \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [K_{EL, \Pi 4}] = k_4 \Delta l_4 \begin{bmatrix} 0 & 0 & 0 \\ 0 & -R \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[K_{s1}] = m_1 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & R \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [K_{s2}] = m_2 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -R \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [K_{s3}] = m_3 g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -R \sin \theta_0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$\sum F_x = 0 \rightarrow T = m_1 \ddot{x}_{c1} + m_2 \ddot{x}_{c2} + m_3 \ddot{x}_{c3} - F_0 \cos \beta$$

$$\sum F_y = 0 \rightarrow N = \sum_{i=1}^3 m_i g + F_{el\phi 1} - F_{el\phi 2} - F_{el\phi 3} + F_{el\phi 4} + \sum_{i=1}^3 m_i \ddot{y}_{ci} - F_0 \sin \beta + F_{el1} - F_{el2} - F_{el3} + F_{el4}$$

$$\Rightarrow N_{ST} = \sum_{i=1}^3 m_i g + F_{el\phi 1} - F_{el\phi 2} - F_{el\phi 3} + F_{el\phi 4}$$

$$N_{dyn} = \sum_{i=1}^3 m_i \ddot{y}_{ci} + F_{el1} - F_{el2} - F_{el3} + F_{el4} - F_0 \sin \beta$$

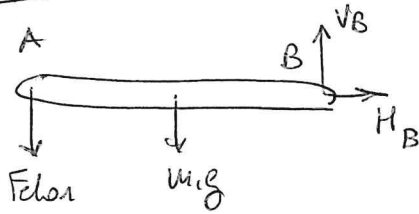
→ question #6: output of the FRF is T and Ndyn

→ question #7: plot time history of T and Ndyn or N
(it was not specified whether Ndyn or N)

a) compute $f_s = \frac{|T|}{N}$ ~~plot~~ as a function of time and get the maximum value

$$b) f_{smin} = \frac{\max(|T|)}{\min(N)}$$

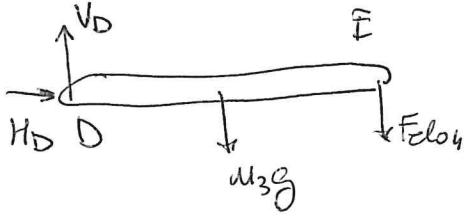
question #8



$$\sum F_x = 0 \rightarrow H_B = 0$$

$$\sum F_y = 0 \rightarrow V_B = w_1 g + F_{\phi 1}$$

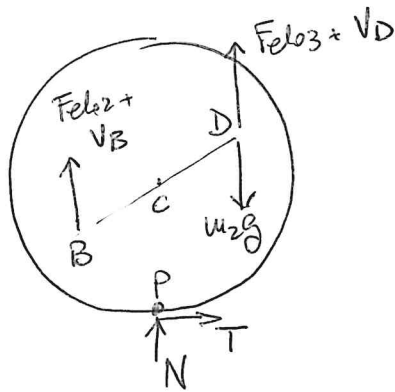
$$\sum M_B = 0 \rightarrow F_{\phi 1} = -\frac{w_1 g}{2}$$



$$\sum F_x = 0 \rightarrow H_D = 0$$

$$\sum F_y = 0 \rightarrow V_D = w_3 g + F_{\phi 4}$$

$$\sum M_D = 0 \rightarrow F_{\phi 4} = -w_3 g/2$$



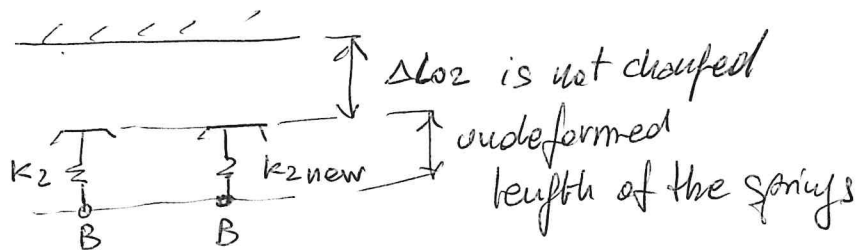
$$\sum M_P = 0 : (F_{\phi 3} + V_D - w_2 g) R \cos \theta_0 = (F_{\phi 2} + V_B) R \cos \theta_0$$

↓

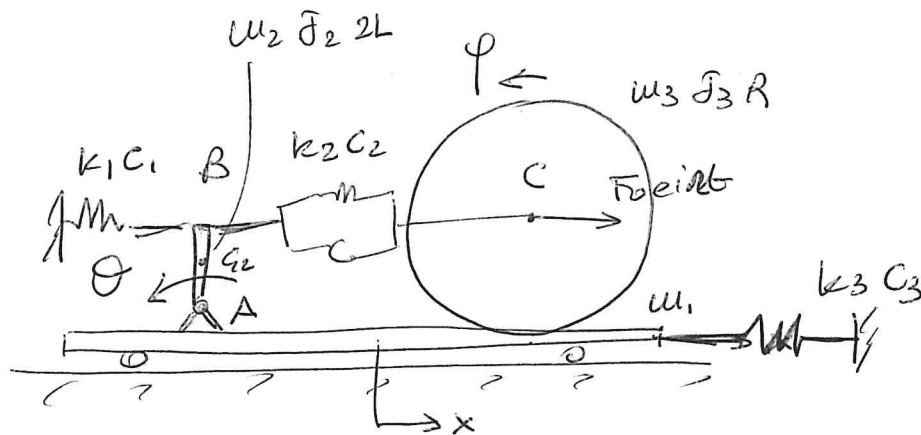
$$F_{\phi 3} = F_{\phi 2} - \frac{w_1 g}{2} + w_2 g + \frac{w_3 g}{2}$$

$$F_{\phi 2_new} = k_{2n} \cdot \Delta \phi_2$$

$$\Delta \phi_{3n} = \frac{F_{\phi 3}}{k_3}$$



- simplified problem -



- 3 rigid bodies $\rightarrow$ 9 dof
- 2 carts, 1 hing + rolling without sliding 6 doc $\rightarrow$ 3 dof

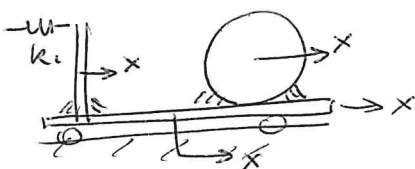
$$\underline{x} = \{x \quad \theta \quad \phi\}^T \quad \underline{x}_0 = \{0 \quad 0 \quad 0\}^T$$

$$T = \frac{1}{2} m_1 \dot{x}^2 + \frac{1}{2} m_2 \dot{x}_{g2}^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_c^2 + \frac{1}{2} J_3 \dot{\phi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^3 c_i \Delta l_i^2 \quad V = \frac{1}{2} \sum_{i=1}^3 k_i \Delta l_i + m_2 g h_{g2} \quad (h_{g1} \text{ and } h_{g3} \text{ are constant})$$

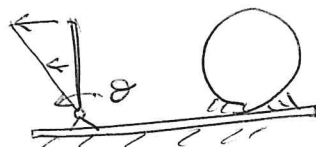
$$\delta W = F_{0 \text{ int}} \delta x_c$$

$$x \neq 0; \theta = \phi = 0$$



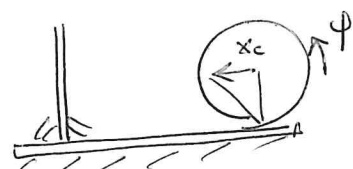
$$\begin{aligned} x_{g2} &= x \\ x_c &= x \\ \Delta l_1 &= x \\ \Delta l_2 &= 0 \\ \Delta l_3 &= -x \end{aligned}$$

$$x = \phi = 0; \theta \neq 0$$



$$\begin{aligned} x_{g2} &= -L_2 \theta \\ x_c &= 0 \\ \Delta l_1 &= -2L\theta \\ \Delta l_2 &= 2L\theta \\ \Delta l_3 &= 0 \end{aligned}$$

$$x = \theta = 0; \phi \neq 0$$



$$\begin{aligned} x_{g2} &= 0 \\ x_c &= -R\phi \\ \Delta l_1 &= 0 \\ \Delta l_2 &= -R\phi \\ \Delta l_3 &= 0 \end{aligned}$$

$$\rightarrow \begin{cases} x_{g2} = x - L\theta \\ x_c = x - R\phi \\ \Delta l_1 = x - 2L\theta \\ \Delta l_2 = 2L\theta - R\phi \\ \Delta l_3 = -x \end{cases}$$

$$M_F = \text{diag}(m_1, m_2, J_2, m_3, J_3)$$

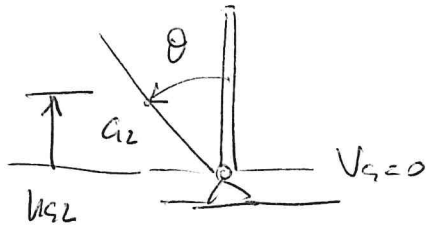
$$C_F = \text{diag}(c_1, c_2, c_3)$$

$$K_F = \text{diag}(k_1, k_2, k_3)$$

$$F = \{f_0\}$$

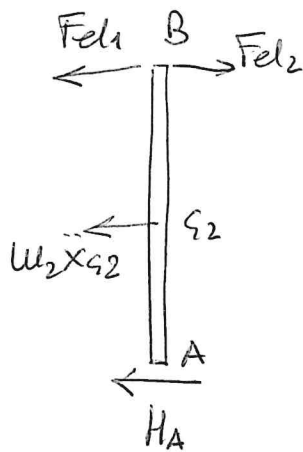
$$[\Delta_{im}] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -L & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -R \\ 0 & 0 & 1 \end{bmatrix} \quad [\Delta_k] = [\Delta_c] = \begin{bmatrix} 1 & -2L & 0 \\ 0 & 2L & -R \\ -1 & 0 & 0 \end{bmatrix}$$

$$[\Delta_e] = \begin{bmatrix} 1 & 0 & -R \end{bmatrix}$$



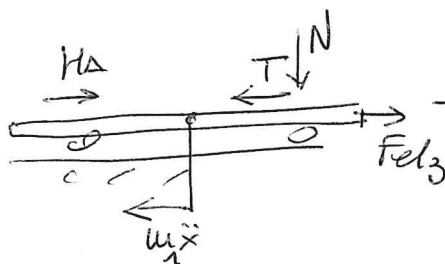
$$u_{s2} = L \cos \theta \rightarrow \frac{\partial^2 u_{s2}}{\partial \theta^2} = -L \cos \theta$$

$$[k_{s2}] = u_{s2} g \begin{bmatrix} 0 & 0 & 0 \\ 0 & -L & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

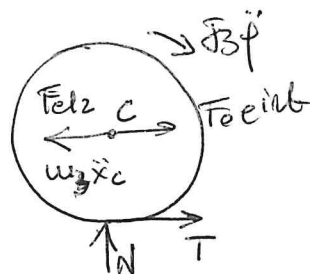


$$\rightarrow \sum F_x = 0 : H_A = F_{el2} - F_{el1} - u_{s2} \ddot{x}_{s2}$$

or:



$$\rightarrow \sum F_x = 0 : H_A = u_{s1} \ddot{x} + T - F_{el3}$$



$$\rightarrow \sum M_c = 0 : T = \frac{J_3 \ddot{\theta}_3}{R}$$